

SPHERE: MISSION CONTROL

STUDENT USER MANUAL & DIGITAL TWIN PROTOCOL

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Inquiry-Based Science Education (IBSE) Focus: This manual provides a complete hardware guide and digital workflow to construct a physical thermal envelope (an astronaut spacesuit capsule) and map its real cooling dynamics inside a vacuum analog to a live Newtonian mathematical prediction engine.

1. INTRODUCTION & THE DIGITAL TWIN PARADIGM

During deep-space voyages, spacecraft and astronauts are subjected to thermal extremes. In low Earth orbit (such as on the International Space Station), direct solar irradiation can heat surfaces to **121°C**, while objects in orbital shadows radiate energy away, plunging to **-157°C**. Creating reliable, multi-layer thermal envelopes is critical to preserving astronaut safety.

In this laboratory experiment, you will build a protective "spacesuit" for a simulated astronaut (a 100ml PET container filled with hot water at 80°C). You will drop this unit into an ice-bath chamber (0°C) mimicking the thermal void of space, and use the **SPHERE: Mission Control** dashboard as your live telemetry terminal.

This project leverages the **Digital Twin Ecosystem**, aligned with the real-world **European Union Horizon 2020 SPHERE initiative**. A Digital Twin is a real-time mathematical replication of a physical device. By pairing your physical sensor readings with the simulator's theoretical predictions, you will evaluate heat dissipation efficiency and insulation integrity!

2. HARDWARE ASSEMBLY GUIDE

To ensure the experiment is economically scalable across diverse classrooms, the hardware relies exclusively on affordable, off-the-shelf industrial and household components:

- **Capsule Body:** 1x 100ml wide-mouth PET food jar.
- **Waterproof Routing:** 1x Industrial PG7 Nylon Cable Gland (mounted securely through a 12.5mm hole drilled in the PET jar cap).
- **Thermal Sensor:** 1x DS18B20 digital waterproof temperature sensor probe.
- **Environmental Chamber:** A large tub or bucket filled with crushed ice and water (maintained at 0°C).

ASSEMBLY STEPS:

1. Thread the digital DS18B20 sensor wire through the PG7 nylon gland body.
2. Screw the PG7 gland locknut onto the inside of the jar cap until firmly sealed.
3. Tighten the external dome nut of the PG7 gland. This compresses the internal neoprene rubber seal around the sensor wire, creating a 100% waterproof seal capable of sustaining submersion in ice water.
4. Verify that the sensor probe is suspended freely in the middle of the jar and does not directly touch the bottom or sides of the PET capsule.

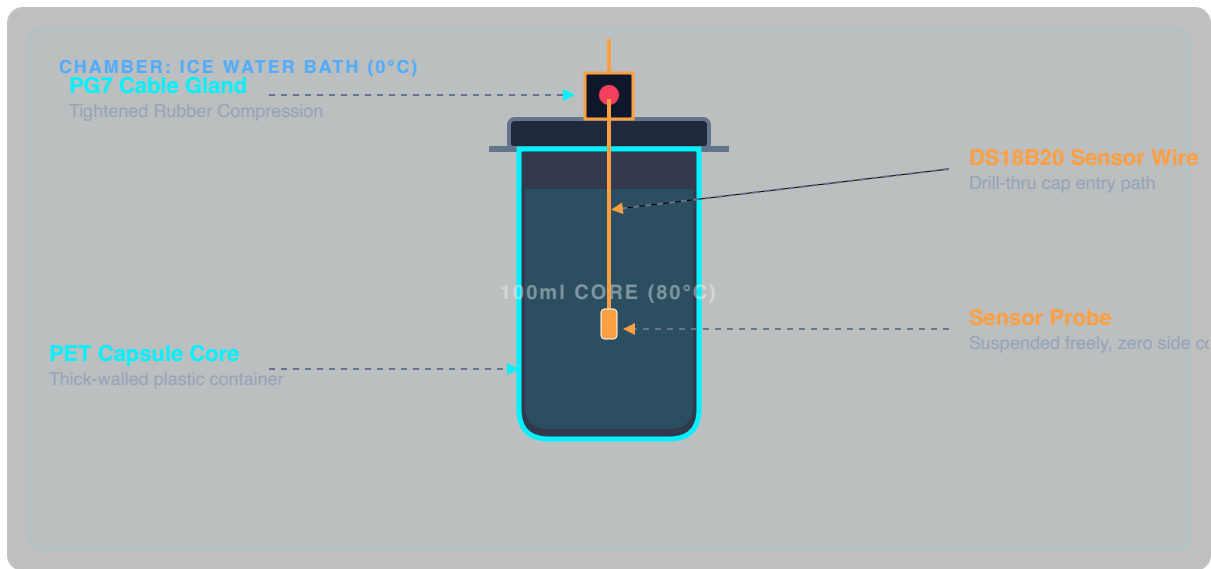


FIGURE 1: SPHERE CAPSULE MECHANICAL ASSEMBLY SCHEMATIC

3. INSULATION SCHEMES & OPTIMIZATION

Your main engineering challenge is to test different insulation configurations wrapped around the physical capsule to slow the temperature decay:

- **Configuration A (Bare Capsule):** No insulation applied. Serves as your scientific control baseline.
- **Configuration B (Cotton Padding):** Represents conduction protection. Heavy cotton layers absorb mechanical heat but allow air movement.
- **Configuration C (Bubble Wrap Rings):** Represents convection prevention. Captures static air pockets, slowing convective heat transfer.
- **Configuration D (Multi-Layer Insulation - MLI):** Reflective aluminum Mylar foil layers separated by thin netting. The gold standard for space flight, designed to block radiant heat transfer.

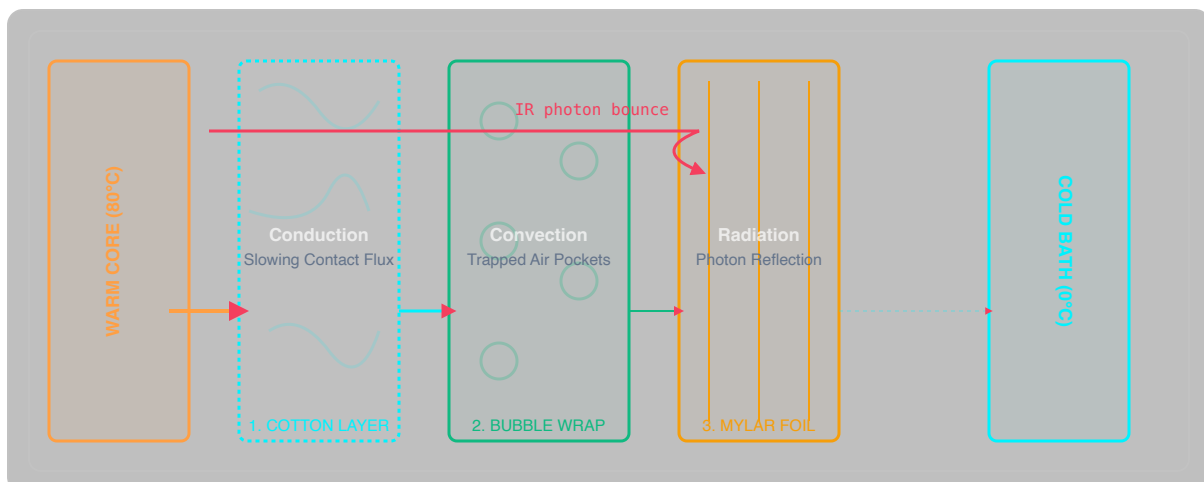


FIGURE 2: THERMAL INSULATION FLUX SCHEMES & BARRIERS

4. MATHEMATICAL CORE & NEWTON'S COOLING FORMULA

The rate at which the warm water capsule cools in the cold ice bath is modeled by **Newton's Law of Cooling**. This law states that the rate of heat loss of an object is directly proportional to the difference in temperatures between the object and its surrounding environment:

$$dT/dt = -k \cdot (T - T_{env})$$

By integrating this differential equation, we get the analytical prediction formula utilized by the **SPHERE Mathematical Engine**:

$$T(t) = T_{env} + (T_0 - T_{env}) \cdot e^{-k \cdot t}$$

Where:

- **T(t)**: The predicted temperature at time **t** (seconds).
- **T_{env}**: The environmental ice-water temperature (fixed at **0°C**).
- **T₀**: The initial temperature of the warm water at drop-time (typically **80°C**).
- **k**: The cooling decay coefficient (1/seconds) which represents the heat transfer rate. A lower **k** constant indicates a more efficient spacesuit!
- **e**: Euler's mathematical constant (~2.71828).

5. STEP-BY-STEP MISSION CONTROL WORKFLOW

1. **Unlock Operations:** Read the Mission Briefing on the home page and complete the Pre-Flight Quiz to prove your laboratory readiness.
2. **Run Newtonian Predictions:** Open the **Simulator** tab. Select your insulation type, configure your initial variables (**T₀ = 80°C**, **T_{env} = 0°C**), and click **RUN SIMULATION PREDICTION**. Review the projected mathematical decay curve.
3. **Prepare Physical Asset:** Fill the PET capsule with hot water pre-measured at 80°C. Secure the cap and tighten the PG7 gland to prevent convective leakage.
4. **Submerge & Initialize:** Submerge the capsule into the 0°C ice water. Simultaneously click **START LIVE TIMER** in the **Live Telemetry** tab.

5. **Active Sensor Logging:** Every 60 seconds (the alarm terminal will blink green and sound a beep), read the current physical thermometer value and log it into the input box. Click **RECORD DATA POINT** to plot it on the graph overlay.
6. **Analyze Correlation Score:** Once at least 5 points are logged, the telemetry panel automatically calculates your **Thermal Correlation Score**. Discuss with your crew how closely your digital twin simulator aligned with physical reality!